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July 1, 2021

Attn: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, DC 20555-0001

10 CFR 50.73

**SUSQUEHANNA STEAM ELECTRIC STATION
LICENSEE EVENT REPORT 50-388/2020-002-01
UNIT 2 LICENSE NO. NPF-22
PLA-7917**

Docket No. 50-388

Attached is Licensee Event Report (LER) 50-388/2020-002-01. The LER supplement reports an event involving drifting of Reactor Pressure Steam Dome – Low permissive switches (Microswitch 2). This event was determined to be reportable as a condition prohibited by Technical Specifications in accordance with 10 CFR 50.73(a)(2)(i)(B), a condition that could have prevented fulfillment of a safety function in accordance with 10 CFR 50.73(a)(2)(v)(D), and a common cause inoperability of independent trains or channels in accordance with 10 CFR 50.73(a)(2)(vii).

There were no actual consequences to the health and safety of the public as a result of this event.

This letter contains no new or revised regulatory commitments.

A handwritten signature in black ink, appearing to read "K. Cimorelli", written in a cursive style.

K. Cimorelli

Attachment: LER 50-388/2020-002-01

Copy: NRC Region I
Ms. A. Klett, NRC Project Manager
Mr. C. Highley, NRC Senior Resident Inspector
Mr. M. Shields, PA DEP/BRP



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form <https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk all: omb_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Susquehanna Steam Electric Station Unit 2

2. Docket Number

05000388

3. Page

1 of 3

4. Title

Condition Prohibited by Technical Specifications Due to Drift of Reactor Pressure Switch Caused by Lack of Requirements for Acclimation of the Instrument to the Operating Environment

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
08	10	2020	2020	- 002 -	01	07	01	2021	Facility Name	Docket Number
										05000

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☒ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ Other (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

C. E. Manges, Jr, Principle Engineer – Nuclear Regulatory Affairs

Phone Number (Include Area Code)

570-542-3089

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
E	JE	PS	GE/Cameron	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On August 10, 2020, the Unit 2 "D" Reactor Steam Dome Pressure – Low permissive pressure switch, Microswitch 2, was found outside of the Technical Specification (TS) 3.3.5.1 allowable value. The switch drifted outside of the upper allowable value which is intended to ensure that the reactor dome pressure has fallen to a value below the Core Spray and Residual Heat Removal (RHR)/Low Pressure Coolant Injection (LPCI) maximum design pressures to preclude over-pressurization of the low pressure systems prior to low pressure injection initiation.

Based on the information available, the condition existed for longer than allowed by TS 3.3.5.1, TS 3.5.1, and TS 3.0.3. As such, this is a condition prohibited by TS and is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B). In addition, since the "C" channel (PIS-B21-2N021C) was surveillance tested just prior to identification of the drift of the "D" channel (PS-B21-2N021D), redundant channels were inoperable at the same time impacting both Core Spray and LPCI functions; therefore, this is also considered a condition that could have prevented fulfillment of a safety function (10 CFR 50.73(a)(2)(v)(D)) and a common cause inoperability of independent trains or channels (10 CFR 50.73(a)(2)(vii)).

Drifting outside of the TS allowable value on August 10, 2020 is attributed to vendor documents not specifying requirements related to soaking, acclimating, or exposing the instrument to the environment prior to installation and the switch not meeting vendor published repeatability of 0.25% of full scale. Key corrective actions include changing the Susquehanna vendor manual associated with the pressure transmitters to identify the need to acclimate new (or refurbished) switches to in service conditions for at least 48 hours prior to placing in service and calibrating.

There were no actual consequences to the health and safety of the public as a result of this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Susquehanna Steam Electric Station Unit 2	05000-388	2020	- 002 -	01

NARRATIVE**CONDITIONS PRIOR TO EVENT**

Unit 1 – Mode 1, approximately 100 percent Rated Thermal Power

Unit 2 – Mode 1, approximately 100 percent Rated Thermal Power

There were no structures, systems, or components that were inoperable at the start of the event that contributed to the event.

EVENT DESCRIPTION

Prior to September 2017, Susquehanna Steam Electric Station (SSES) had been utilizing International Telephone and Telegraph (ITT)-Barton 288A pressure switches in the Reactor Steam Dome Pressure – Low channels [EIS System/Component Identifier: JE/PS] that provide the injection permissive for the Core Spray system [EIS System Identifier: BM] (Technical Specification (TS) 3.3.5.1, Function 1d) and the Residual Heat Removal (RHR)/Low Pressure Coolant Injection system (LPCI) [EIS System Identifier: BO] (TS 3.3.5.1, Function 2d). Due to instrument drift concerns, all eight obsolete ITT-Barton 288A pressure switches were replaced with General Electric (GE) recommended Cameron-Barton 288A pressure switches between September 6, 2017 and November 15, 2017. Following replacement, instrument drift continued to be an issue, and additional corrective action was determined necessary to resolve the drift concerns. The additional corrective action included procuring and installing Cameron-Barton 288A instruments that had been modified to remove an over-range condition and the movement assembly/associated linkages that were determined to be affecting instrument drift. The first of these modified instruments was installed and calibrated on July 8, 2020 in the Unit 2 “D” channel (pressure switch for PS-B21-2N021D).

On August 10, 2020, the new PS-B21-2N021D switch, Microswitch 2, was tested for its first surveillance interval and found to be outside of TS acceptance criteria. The switch drifted 1.2 psig outside of the upper allowable value which is intended to ensure that the reactor dome pressure has fallen to a value below the Core Spray and RHR/LPCI maximum design pressures to preclude over-pressurization of the low pressure systems prior to low pressure injection initiation.

Based on the information available, the condition existed for longer than allowed by TS 3.3.5.1, “Emergency Core Cooling System (ECCS) Instrumentation,” TS 3.5.1, “ECCS-Operating,” and TS 3.0.3. As such, this is a condition prohibited by TS and is being reported in accordance with 10 CFR 50.73(a)(2)(i)(B). In addition, since the “C” channel (PIS-B21-2N021C) was surveillance tested just prior to identification of the drift of the “D” channel (PS-B21-2N021D), redundant channels were inoperable at the same time impacting both Core Spray and LPCI functions; therefore, this is also considered a condition that could have prevented fulfillment of a safety function (10 CFR 50.73(a)(2)(v)(D)) and a common cause inoperability of independent trains or channels (10 CFR 50.73(a)(2)(vii)).

CAUSE OF EVENT

Drifting outside of the TS allowable value on August 10, 2020 is attributed to vendor documents not specifying requirements related to soaking, acclimating, or exposing the instrument to the environment prior to installation and the switch not meeting vendor published repeatability of 0.25% of full scale.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Susquehanna Steam Electric Station Unit 2	05000-388	2020	- 002 -	01

NARRATIVE**ANALYSIS/SAFETY SIGNIFICANCE**

All components in the LPCI and Core Spray would have been able to withstand a pressure of 456.2 psig (434.2 psig with water head removed which allows direct comparison to TS value) which was the maximum recorded pressure for instrument PS-B21-2N021D. Therefore, Core Spray and RHR would have been able to perform its safety function and the condition described herein did not result in a safety system functional failure. Accordingly, this event will not be counted as a safety system functional failure in the Reactor Oversight Process Performance Indicators. There were no actual consequences to the health and safety of the public as a result of this event.

CORRECTIVE ACTIONS

The switch was acclimated and returned to within the TS allowable values. Key corrective actions include the following:

1. Changed the Susquehanna vendor manual associated with the pressure transmitters to identify that acclimation of new (or refurbished) switches to in service conditions for at least 48 hours prior to placing in service and calibrating is required.

COMPONENT FAILURE INFORMATION

The switches installed in the application affected by this condition are Cameron-Barton 288A pressure indicating switches manufactured by Cameron. The switch that drifted in the event described in this LER had been modified to remove an over-range condition and the movement assembly/associated linkages; with the indication function removed, this switch was a pressure switch.

PREVIOUS OCCURRENCES

As indicated by the previous occurrences listed below, the condition of drifting pressure indicating switches is a known issue at SSES. Previously reported Corrective Actions are in progress; additional corrective actions will be dictated by the Susquehanna Corrective Action Program.

LER 50-387(388)/2018-005-01, "Condition Prohibited by Technical Specifications Due to Drift of Reactor Pressure Switches", dated December 16, 2019.

LER 50-388/2017-010-01, "Condition Prohibited by Technical Specifications Due to Drift of Reactor Pressure Switches", dated December 16, 2019.

LER 50-388(387)/2015-001-01, "Condition Prohibited by Technical Specifications Due to Drift of Reactor Pressure Steam Dome-Low Switches", dated February 10, 2016.